Advancing Healthcare With Large Language Models: Techniques and Application

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***Abstract*—As the costs of global healthcare systems continue to rise, large language models (LLMs) have emerged as a promising technology with vast potential and wide-ranging applications in the medical field. We provide a detailed overview of the lifecycle of medical LLMs, encompassing three key stages: get, refine, and use, aimed at assisting healthcare practitioners and patients in utilizing these models more effectively. We also summarize the currently widely used medical evaluation benchmarks, analyzing their advantages and limitations. Furthermore, we conduct a comparative analysis of specialized medical LLMs on benchmarks such as MedQA, PubMedQA, MMLU-MED, and MedMCQA, revealing that methods like retrieval-augmented generation (RAG) enable smaller models to outperform larger ones by effectively integrating external medical knowledge. This review provides a reference for medical professionals to evaluate LLM capabilities and inspires the development of more effective benchmarking methods. Additionally, we showcase practical applications of medical LLMs in clinical, research, and educational settings, providing healthcare workers with valuable resources. Finally, we identify current challenges faced by medical LLMs and present outlooks for future technological advancements, aiming to inspire users to explore new ways to address existing issues. This review serves as an entry point for interested clinicians, helping them determine whether and how to integrate LLM technology into healthcare for the benefit of patients and practitioners.**

***Index Terms*—Application, healthcare, large language models (LLMs), medicine, techniques.**

I. Introduction

THE COVID-19 pandemic has dramatically exposed the limitations of traditional healthcare systems, which struggled with a surge in patient numbers during global health crises [1]. This sudden influx stressed healthcare infrastructures, leading to extended waiting times, heightened infection risks, and significant increases in administrative burdens [2]. Such challenges intensified stress among healthcare professionals and worsened patient outcomes, highlighting deep-seated issues like diagnostic errors, inefficiencies, and unequal resource distribution [3]. Moreover, the vast volume and complexity of medical knowledge required in these times have overwhelmed healthcare workers, making it difficult to stay updated with the latest information [4]. These conditions underscore the critical need for innovative technological solutions that enhance healthcare efficiency and support effective clinical decision-making.

To address these pressing healthcare system crises, innovative solutions such as large language models (LLMs) are urgently needed. LLMs, a category of artificial intelligence that leverages deep learning techniques, are trained on vast corpora, enabling them to generate coherent and realistic text that closely mimics human language [5]. The release of ChatGPT by OpenAI in November 2022, an LLM based on transformer neural network technology [6], marked a significant milestone, showcasing exceptional language comprehension capabilities within the medical field. This development has not only introduced innovative tools into medical practice but also heralded a new era in medical technology.

LLMs engage directly with patients to extract essential medical information, support physicians in the diagnostic process, and save valuable consultation time [7]. These models address patients' informational needs about their conditions [8], helping to alleviate anxiety and foster trust in the treatment process. Moreover, LLMs assist healthcare professionals in summarizing medical literature and managing documentation, thereby reducing the burden of research and administrative tasks [9]. Ayers *et al.* [10] suggest that patients often prefer the empathetic responses provided by LLMs over traditional interactions, as these responses are perceived as more caring and personalized, thus enhancing the overall healthcare experience.

Given these advancements, a deeper understanding and continued exploration of LLM applications in healthcare are not merely beneficial; they represent an inevitable progression in medical technology. Our contributions to this evolving field are significant and multifaceted:

1) Innovative medical LLM lifecycle framework. To the best of our knowledge, it is the first systematic review of the

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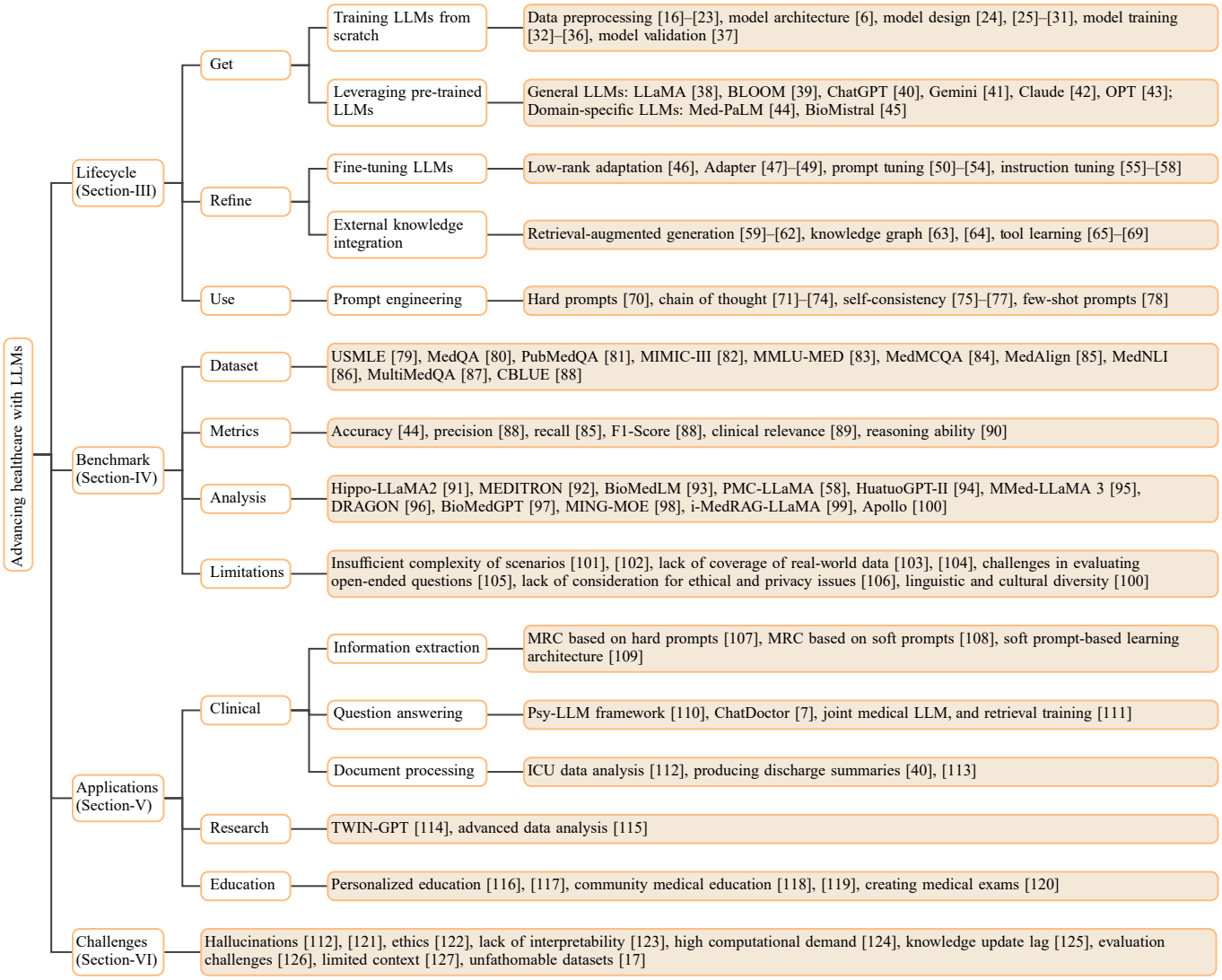


Fig. 1. Advancing healthcare with large language models: techniques and applications.

complete lifecycle of LLMs in the medical domain, including three critical stages: get, refine, and use. This framework provides healthcare professionals with a clear path to understanding and implementing these powerful technological tools.

2) Comprehensive evaluation metrics and methodological critique. We conducted a comparative analysis of existing models on widely-used medical benchmarks, highlighting the strengths and weaknesses of various methodologies and introducing a novel perspective for evaluating medical LLMs. Our experiments demonstrate that effective refinement enables smaller models to outperform larger ones. Additionally, we discuss the limitations of current evaluation benchmarks. This contribution underscores the importance of robust evaluation processes and offers empirical insights to inform more precise frameworks for future research.

3) In-depth applications and challenges analysis with proposed solutions. We conduct a thorough examination of how medical LLMs are implemented across various sectors of healthcare, including clinical practice, research, and education. Our analysis highlights the innovative solutions offered by these technologies and identifies persistent challenges that limit their effectiveness. We propose viable solutions to these issues, aiming to enhance the operational capabilities of medical LLMs and improve healthcare outcomes.

Existing reviews on LLMs in healthcare often focus on specific aspects, resulting in coverage that lacks comprehensiveness. For example, Haltaufderheide and Ranisch [11] restrict their discussion to the ethical implications of LLMs in healthcare, while Neveditsin *et al.* [12] and Yu *et al.* [13] limit their focus to clinical applications and development in biomedical practices without adequately addressing operational methods for practical implementation. Similarly, discussions by Guo *et al.* [14] and Lawrence *et al.* [15] are confined to mental health, omitting broader applications across medical scenarios.

In contrast, this paper comprehensively discusses the lifecycle of medical LLMs, and provides detailed analyses of evaluation benchmarks, application scenarios, as well as current challenges and future perspectives. Our review aims to serve as a comprehensive resource for global healthcare practitioners and researchers, assisting them in understanding and strategically leveraging medical LLMs to drive innovation and optimization in healthcare services. The structure of this paper is illustrated in Fig. 1.

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II. Background

The development of LLMs marks a pivotal advancement in natural language processing (NLP), initiated by transformative models like bidirectional encoder representations from transformers (BERT) [128] and generative pretrained transformer (GPT) [129]. These models, rooted in the Transformer architecture introduced in 2017 [6], have revolutionized our understanding and generation of human-like text. Unlike earlier models that processed words sequentially, Transformers handle words in parallel while maintaining a comprehensive view of the entire text, significantly enhancing coherence and context-awareness in language processing [24].

As LLM technology has evolved, its potential has extended beyond general NLP tasks into specialized domains such as healthcare, where the capacity to process both structured and unstructured data is crucial. The healthcare sector, with its complex data comprising medical literature, patient records, and diverse health databases, has emerged as a prime field for applying advanced NLP tools [90]. This integration has been driven by the growing need to efficiently manage, interpret, and utilize vast data volumes to derive meaningful insights and support medical decision-making [40], [120].

The shift of LLMs into healthcare has paralleled the broader digital transformation within the industry. The adoption of electronic health records (EHRs), digital patient engagement platforms, and AI-driven diagnostic tools is a notable example of how LLMs are being employed to enhance service capabilities. These models offer round-the-clock question-answering (QA) services, provide real-time, personalized responses to patient inquiries, and support continuous education for medical professionals through personalized learning tools [45], [89].

Kung *et al.* [79] evaluated the performance of ChatGPT on the United States medical licensing examination (USMLE), showing that even without specialized medical training, this model achieved scores near the 60% passing threshold. Building upon this foundation, the subsequent version, GPT-4, shows marked improvements in medical application accuracy, with scores increasing to 90% on the USMLE, highlighting the benefits of iterative model refinement and specialized training in healthcare applications [130]. This indicates that general LLMs like ChatGPT can competently handle tasks that require substantial clinical insight, providing foundational support for medical education and preliminary diagnostics.

In contrast to general-purpose models, MedPaLM [87]—a model specifically trained with medical data—demonstrates even more robust performance in the medical domain. It was the first LLM to surpass the "passing" score in USMLE-style questions, achieving an impressive 86.5% accuracy and scoring 67.2% on the MedQA dataset [80]. This superior performance underscores the significant impact of domain-specific training on the effectiveness of LLMs in clinical settings. MedPaLM 2, further trained with targeted medical knowledge, continued this trend by showing exceptional ability in predicting depression scores and other complex medical tasks, thereby confirming the vast potential of specialized LLMs in enhancing the accuracy and efficiency of healthcare services [131].

However, the application of LLMs in medical settings is not without challenges. Their performance varies significantly based on the complexity of the task and the specificity of the training data. For example, in assessments like the MRCOG exams, ChatGPT demonstrated a competent grasp on foundational medical knowledge but struggled with more complex clinical reasoning required in advanced stages [132]. This discrepancy underscores the need for continuous advancements in model training and specialization to address complex clinical scenarios effectively.

III. The Lifecycle of Medical LLMs

As LLMs become increasingly integral to the medical field, understanding their lifecycle is essential for effective deployment and utilization. The lifecycle of medical LLMs involves three critical stages: getting the model, refining the model, and using the model. Each stage entails specific technical and practical steps to ensure the model operates efficiently, safely, and reliably within healthcare environments.

A. How to Get Medical LLMs

In modern healthcare, LLMs are increasingly being used to improve the quality, efficiency, and personalization of medical services, assisting healthcare professionals with tasks such as medical record analysis and disease diagnosis [133]. However, when medical institutions decide to adopt these models, they face a choice between two main approaches: Training LLMs from scratch or leveraging pre-trained LLMs.

1) Training LLMs from Scratch: Training a LLM from scratch allows healthcare institutions to have complete control over the entire development process, enabling them to customize each step to ensure that the model aligns closely with their specific needs. This is particularly beneficial when dealing with specialized medical data, such as records of specific diseases or clinical data from particular patient groups. Custom training can significantly improve the model's performance and practicality, reducing the risk of misdiagnoses or incorrect recommendations while also avoiding data privacy concerns associated with external data sharing [134].

a) Data Preprocessing: Data preprocessing is the first critical step in training a LLM from scratch. It ensures that the input data provided to the model is high-quality, accurate, and suitable for training. In the medical field, the quality of data preprocessing directly impacts the model's performance and real-world application [16]. Therefore, the goal of data preprocessing is to eliminate noise, fill in missing values, and standardize data formats to create a consistent and clean medical training dataset [17].

i) Data Collection: Data collection forms the foundation for training LLMs, determining the scope and depth of the model's knowledge [18]. In the medical field, the diversity and scale of data are crucial, directly influencing the model's generalizability and adaptability [19]. Effective data collection not only considers the quantity of data but also its quality, the legality of the sources, and its relevance to model training [45]. Different healthcare institutions, patient groups, and even geograph-

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ical regions may exhibit different medical practices, terminologies, and disease presentations [135]. Therefore, collecting high-quality medical data that covers a wide range of cases, diseases, treatment methods, and patient backgrounds is essential. Bi *et al.* [136] proposed a law on the scaling of models and data, introducing a new model scale representation in non-embedding FLOPs per token M, replacing the earlier-used model parameters N. They substituted the approximate compute budget formula $C = 6ND$ with a more precise formula $C = MD$. To determine the optimal model size $M_{\text{opt}}(C)$ and data size $D_{\text{opt}}(C)$, they formulated the problem as minimizing the loss function $L(N, D)$ subject to the constraint $C = M \times D$:

$$(M_{\text{opt}}(C), D_{\text{opt}}(C)) = \arg \min_{\substack{M,D \\ C=M\times D}} L(N, D). \tag{1}$$

This optimization ensures that the model and data sizes are chosen to minimize the loss while adhering to the computational budget. Bi *et al.* [136] also found that high-quality data can drive the training of larger models even with the same data size, emphasizing the importance of data quality in model performance.

Hospitals can use various data types when training LLMs, including medical literature, EHRs, clinical notes, diagnostic reports, and more [87], [137]. In addition to using their own datasets, hospitals can also leverage publicly available datasets, such as MedQuAD [138], RJUA-QA [102], and MedRedQA [104], to enhance the learning and application capabilities of the model.

ii) Data Cleaning: In the medical field, effective cleaning and preprocessing of collected raw data are crucial to ensure data quality and consistency, thereby supporting accurate medical decision-making and research [20]. Data cleaning involves removing noise, such as spelling errors and duplicate records [139], and using spell-checking and deduplication algorithms to ensure data accuracy and dataset balance [140]. Additionally, missing data can be handled through deletion or data imputation methods [21]. Moreover, data should be standardized and normalized, particularly for medical indicators such as blood pressure and blood glucose levels, to eliminate dimensional differences and ensure data consistency [141].

To accommodate the multivariate nature of medical data, the standardization process can be mathematically represented as:

$$\mathbf{Z} = \frac{\mathbf{X} - \boldsymbol{\mu}}{\boldsymbol{\sigma}} \tag{2}$$

where $\mathbf{X}$ is the original data matrix with n samples and m features, $\boldsymbol{\mu}$ is the mean vector of the features, and $\boldsymbol{\sigma}$ is the standard deviation vector for each feature. This vectorized approach ensures that each feature is standardized independently, maintaining the relationships between different medical indicators while ensuring data consistency across the dataset. These steps significantly improve the performance and accuracy of machine learning models in medical data analysis.

iii) Feature Extraction: After data cleaning, medical data typically still contain a large number of variables and information, but not all of these data are practically helpful for model training [22]. Therefore, it is necessary to identify and extract the most informative and predictive features through feature extraction. Important feature identification uses methods such as recursive feature elimination (RFE) [142], linear discriminant analysis (LDA) [143], principal component analysis (PCA) [144], and random forest feature selection [142] to help identify variables with the highest predictive value for model training while removing redundant or irrelevant variables. Choubey *et al.* [145] demonstrated that both PCA and LDA methods can effectively identify important features, reduce costs and computational time, and improve accuracy. Additionally, feature engineering involves transforming, combining, or decomposing raw data to generate new useful features [146], such as calculating BMI or combining different blood indicators to improve the accuracy of disease prediction [23]. These steps significantly improve the predictive accuracy and efficiency of the model by reducing data noise and enhancing data quality.

b) Model Architecture: The Transformer architecture, foundational for modern LLMs, is particularly effective for medical applications due to its advanced handling of complex language data [6]. It incorporates a self-attention mechanism that allows the model to prioritize contextually important information across extensive medical documents. Multi-head attention further enhances this by enabling simultaneous processing of different linguistic elements, crucial for analyzing the multifaceted nature of medical data. Additionally, positional encoding preserves the sequential integrity of information, essential for accurately interpreting structured data such as patient histories and treatment protocols. Collectively, these features enable medical LLMs to generate reliable, contextually accurate outputs, supporting a wide range of tasks from patient interaction to clinical decision-making.

c) Model Design: The Transformer architecture, pivotal for LLMs, varies across models to suit different NLP tasks: Encoder-only models like BERT [128] are ideal for comprehension tasks due to their bidirectional self-attention, which effectively processes contextual relationships. Decoder-only models such as GPT [25] excel in generative tasks by employing unidirectional self-attention to produce text. Encoder-Decoder models, like T5 [24] and BART [26], handle both comprehension and generation, making them suitable for complex tasks like translation and QA.

Optimizing these architectures for medical applications involves adapting and extending them to manage the nuances of medical data. For instance, increasing encoder layers in BERT enhances its ability to parse detailed clinical records [27], while more decoder layers in GPT models improve their generative capabilities for medical text [28]. Utilizing advanced activation functions like GELU [29] and appropriate initialization methods such as Xavier [31] and He *et al.* [30] ensures the models remain stable and effective throughout training, crucial for the long-term processing of intricate medical information.

d) Model Training: Training is a critical phase in the development of medical LLMs, setting the foundation for their effectiveness in healthcare applications. Essential compo-

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nents include:

i) Training Environment Setup: For LLMs, especially in handling extensive medical datasets, it's crucial to establish a robust training infrastructure. This often involves configuring high-memory GPUs or TPUs and employing distributed training techniques like Horovod [32] to enhance efficiency and scalability.

ii) Optimizer and Loss Function: Effective training of medical LLMs typically employs the Adam optimizer [33] for its adaptability across different data scales, combined with advanced loss functions tailored to the specific requirements of medical data processing. These loss functions address common challenges in medical datasets, such as class imbalance and the need for robust regression performance, by integrating weighted cross-entropy [147] for classification tasks and Huber loss [148] for regression tasks.

The weighted cross-entropy (WCE) loss function can be defined as:

$$\mathcal{L}_{\text{WCE}} = -\sum_{i=1}^{N}\sum_{c=1}^{C}\alpha_c y_{ic}\log(\hat{y}_{ic}) \tag{3}$$

where y_{ic} is the true label for class c of sample i, $\hat{y}_{ic}$ is the predicted probability for class c of sample i, and α_c is the weight assigned to class c to address class imbalance.

For regression tasks, the Huber loss function is used to combine the advantages of both mean squared error and mean absolute error, providing robustness to outliers:

$$\mathcal{L}_{\text{Huber}} = \begin{cases} \frac{1}{2}(y_i - \hat{y}_i)^2, & \text{for } |y_i - \hat{y}_i| \le \delta \\ \delta|y_i - \hat{y}_i| - \frac{1}{2}\delta^2, & \text{otherwise} \end{cases} \tag{4}$$

where y_i is the true value, $\hat{y}_i$ is the predicted value, and δ is a threshold parameter that determines the point where the loss function changes from quadratic to linear.

To facilitate multi-task learning, where the model simultaneously performs classification and regression, a combined loss function is employed:

$$\mathcal{L} = \alpha\mathcal{L}_{\text{WCE}} + \beta\mathcal{L}_{\text{Huber}} \tag{5}$$

where α and β are hyperparameters that balance the contribution of each loss component.

Additionally, to prevent overfitting and enhance generalization, regularization terms such as L2 regularization [149] can be incorporated into the loss function:

$$\mathcal{L}_{\text{total}} = \alpha\mathcal{L}_{\text{WCE}} + \beta\mathcal{L}_{\text{Huber}} + \lambda\|\theta\|_2^2 \tag{6}$$

where λ is the regularization coefficient and θ represents the model parameters.

The Adam optimizer updates parameters θ using the following adaptive learning rate rules:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1)g_t \tag{7}$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2 \tag{8}$$

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t} \tag{9}$$

$$\hat{v}_t = \frac{v_t}{1-\beta_2^t} \tag{10}$$

$$\theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon}\hat{m}_t \tag{11}$$

where g_t is the gradient at time step t, β_1 and β_2 are the decay rates for the moving averages of the gradient and its square, respectively, η is the learning rate, and ϵ is a small constant to prevent division by zero.

These advanced loss functions significantly improve the performance and accuracy of machine learning models in medical data analysis by addressing specific challenges inherent in medical datasets.

iii) Training Strategies: Building medical LLMs involves strategies tailored to the complexity of medical data. Initiatives like self-supervised learning enable models to learn from vast amounts of unlabeled data, which is common in medical datasets. Additionally, techniques such as learning rate warmup and cosine annealing [34] are implemented to stabilize training phases and enhance convergence.

The cosine annealing learning rate schedule can be mathematically expressed as:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})\left(1 + \cos\left(\frac{T_{\text{cur}}}{T_{\max}}\pi\right)\right) \tag{12}$$

where η_t is the learning rate at time step t, $\eta_{\min}$ and $\eta_{\max}$ are the minimum and maximum learning rates, T_{cur} is the number of epochs since the last restart, and $T_{\max}$ is the maximum number of epochs.

iv) Hyperparameter Tuning: To maximize model performance, hyperparameter tuning is conducted during or after the initial training phases. Tools like Optuna [35] and Hyperopt [36] automate the search for optimal parameter configurations, testing different combinations and updating strategies based on performance feedback. This process is crucial for fine-tuning the model to ensure the best outcomes in practical medical applications.

e) Model Validation: Model validation is crucial in assessing the performance and generalization capability of medical LLMs during training. A fundamental technique employed is cross-validation, particularly k-fold cross-validation [37]. In this method, the dataset is divided into k equally sized subsets. The model is repeatedly trained on $k-1$ subsets while the remaining subset serves as the validation set. This cycle is repeated such that each subset is used as the validation set once. This approach maximizes the use of available data, reduces the risk of information leakage, and helps mitigate the potential for model overfitting.

In conclusion, training LLMs from scratch provides healthcare organizations with complete control over the development process, enhancing data privacy and security. Customized models are specifically tailored to the intricate needs of medical tasks, demanding substantial resources and expertise. However, the advantages of such customization—increased flexibility and precise data handling—offer substantial long-

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term benefits, improving the reliability and practical applicability of medical LLMs.

2) Leveraging Pre-trained LLMs: Selecting pre-trained LLMs involves key considerations like domain applicability and the method of model acquisition. Domain applicability impacts the model's effectiveness in specific medical tasks, necessitating a choice between general LLMs and domain-specific LLMs. The acquisition method affects usage flexibility and cost, distinguishing between open-source and closed-source models.

Open-source LLMs, such as LLaMA [38], BLOOM [39], and OPT [43], are freely available and modifiable, suitable for institutions requiring customizable solutions with considerations for data privacy and control over computational resources.

Conversely, closed-source models provided by companies, like ChatGPT [120], Gemini [41], and Claude [42], offer simplified deployment via APIs or cloud services. These reduce hardware setup complexities but may involve higher operational costs and potential data privacy risks, with limited customization capabilities for specific medical tasks.

a) General LLMs: Trained on diverse data, general LLMs like ChatGPT and LLaMA [38] are adaptable across various NLP tasks but may lack depth in specialized medical fields [150]. Their broad training can lead to inherent biases, which are critical to manage in medical applications due to ethical concerns [106].

b) Domain-Specific LLMs: These models are tailored for particular fields, such as Med-PaLM [44] and BioMistral [45], offering detailed knowledge and better performance in specific domains like healthcare. However, they require substantial resources for training on domain-specific data, which could be costly and challenging to update.

Both types of LLMs bring strengths to medical applications, but their limitations must be carefully weighed in light of the intended use, with general LLMs offering broad applicability and domain-specific LLMs delivering targeted expertise. Table I summarizes some common LLMs.

TABLE I
OVERVIEW OF SELECTED LLMS

Name	Parameters (Billion)	Open source	Domain specificity
GPT-1	0.117	×	×
BERT	0.34	√	×
ClinicalBERT	0.11	√	√
BART	0.374	√	×
BioBERT	0.11	√	√
GPT-2	1.5	×	×
GPT-3	175	×	×
GPT-4	1800	×	×
BLOOM	176	√	×
Mistral	7	√	×
BioMistral	7	√	√
PaLM	540	×	×
Med-PaLM	540	×	√
Claude 3	500	×	×
Orion	14	√	×
MEDITRON	70	√	√
LLaMA 3	70	√	×
Baichuan 2	13	√	×
GLM	130	√	×
ClinicalCamel	70	√	√
IvyGPT	33	√	√

B. How to Refine Medical LLMs

In the previous Section III-A, we explored multiple pathways for acquiring medical LLMs, including training from scratch and leveraging pre-trained models. These models provide powerful tools for healthcare institutions to handle complex language tasks. However, further optimization and adjustment are essential to achieve optimal performance in real-world medical applications. This process includes fine-tuning techniques to adapt the model to specific tasks and integrating external knowledge to enhance reasoning and knowledge coverage.

1) Fine-Tuning: Fine-tuning refers to the process of adjusting the parameters of a pre-trained model to tailor it to specific tasks or domain datasets. By fine-tuning medical LLMs, the models can excel in various complex medical tasks, significantly enhancing their application value.

For instance, Sivarajkumar *et al.* [90] assessed LLMs' capability to extract physical rehabilitation information from clinical notes. They tested the models using zero-shot and few-shot methods to identify key concepts in the provided sections. The results showed that while ChatGPT-3.5 exhibited high recall, it had relatively low precision and F1 scores. They also evaluated the BioClinicalBERT [151] model pre-trained on the Medical Information Mart for Intensive Care-III (MIMIC-III) dataset [82], which, due to insufficient data, performed similarly to simpler methods. This underscores the limitations of deploying LLMs for specific domain tasks without fine-tuning and highlights the importance of dataset quality and size in successful fine-tuning outcomes.

Traditional fine-tuning often involves adjusting all parameters of a pre-trained model with a smaller task-specific dataset. Despite the reduced data size, this method still demands significant computational resources as the entire model undergoes retraining. Moreover, each downstream task necessitates a separate fine-tuned model copy, which can be resource-intensive [51]. This approach may lead to inefficiencies, as different tasks utilize the same foundational pre-trained model.

To improve fine-tuning efficiency, parameter-efficient fine-tuning (PEFT) has emerged as a widely applied strategy, which adapts the model to specific tasks by only adjusting a small portion of the pre-trained parameters [125]. The advantage of PEFT lies in its significant reduction in computational resource consumption and memory demand, making fine-tuning feasible even in resource-limited environments.

Next, we will explore several common PEFT methods. The advantages and disadvantages of each method are shown in Table II.

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TABLE II
COMPARISON OF FINE-TUNING METHODS: ADVANTAGES AND DISADVANTAGES

Method	Advantages	Disadvantages	References
LoRA	1) Efficient resource usage 2) Low computational cost 3) Low model storage requirement 4) No need for large task-specific datasets	1) Limited parameter adjustment 2) Moderate dependency on pre-trained models	[46], [152]
Adapter	1) High flexibility for parameter adjustment 2) Adapts flexibly to different tasks 3) Low computational resource demand 4) Low dependency on pre-trained models	1) Moderate training difficulty 2) Moderate model storage requirement 3) High storage cost	[47]–[49], [153]
Prompt tuning	1) Very low computational resource demand 2) Very low model storage requirement 3) Quick adaptation to new tasks	1) High training difficulty 2) Dependent on prompt design 3) Needs large task-specific datasets	[50]–[54], [108]
Instruction tuning	1) Suitable for multi-step reasoning and complex tasks 2) Moderate computational resource demand 3) Enhanced instruction comprehension and execution	1) High training and storage costs 2) Moderate flexibility of parameter adjustment	[55]–[58]

a) Low-Rank Adaptation (LoRA): LoRA [46] is a PEFT technique that introduces low-rank matrices to adapt pre-trained model weights for specific tasks. The core idea of LoRA is to transform the fine-tuning problem into a low-rank matrix learning problem, thereby significantly reducing the computational resources and storage costs required during the fine-tuning process.

LoRA achieves this by decomposing the weight matrix W_0 of the pre-trained model into the product of two low-rank matrices, $A \in \mathbb{R}^{r\times d}$ and $B \in \mathbb{R}^{d\times k}$, where r is the rank much smaller than d and k, allowing for efficient parameter adaptation. Specifically, the adapted weight matrix W is expressed as:

$$W = W_0 + \Delta W = W_0 + A \times B \tag{13}$$

where $\Delta W = A \times B$ represents the low-rank update to the original weights. Here, $W_0 \in \mathbb{R}^{n\times m}$ is the original pre-trained weight matrix, $A \in \mathbb{R}^{n\times r}$, and $B \in \mathbb{R}^{r\times m}$, with $r \ll \min(n,m)$.

The optimization objective in LoRA focuses on minimizing the loss function $\mathcal{L}$ with respect to the low-rank matrices A and B:

$$(A_{\text{opt}}, B_{\text{opt}}) = \arg \min_{\substack{A,B \\ C=M\times D}} \mathcal{L}(W_0 + A \times B) \tag{14}$$

subject to the rank constraint $\text{rank}(A \times B) \le r$. This constraint ensures that the update ΔW remains low-rank, preserving the efficiency gains by limiting the number of additional parameters.

This decomposition leverages the principle of low-rank approximation in linear algebra, which posits that in high-dimensional spaces, many variations can be effectively captured by a few dominant directions. By learning these directions through A and B, LoRA efficiently captures task-specific feature variations without altering the original model weights. During fine-tuning, only the parameters of A and B are updated, which drastically reduces the number of trainable parameters and computational overhead compared to traditional fine-tuning methods.

Moreover, LoRA can be applied to multiple layers within the Transformer architecture, enabling a scalable and flexible adaptation process across different model depths. For instance, in a Transformer layer, LoRA modifies the query and key projections as follows:

$$Q = W_0^Q + \Delta W^Q = W_0^Q + A^Q \times B^Q \tag{15}$$

$$K = W_0^K + \Delta W^K = W_0^K + A^K \times B^K \tag{16}$$

where W_0^Q and W_0^K are the original query and key projection matrices, and A^Q, B^Q and A^K, B^K are the respective low-rank matrices for queries and keys. This layer-specific adaptation allows LoRA to fine-tune the model in a targeted manner, enhancing its performance on specific medical tasks while maintaining overall model efficiency.

LoRA has demonstrated remarkable results on resource-limited medical devices. By adjusting only a few parameters, LoRA enables these devices to operate models that perform comparably to those on larger servers, offering real-time medical support and decision-making assistance. For example, Yu *et al.* [152] demonstrated significant effectiveness in extracting adverse drug event terms by fine-tuning an LLM with LoRA.

b) Adapter: Adapters [47] are a popular PEFT method. They add small, trainable layers to the existing Transformer architecture of the pre-trained model, updating only these layers during fine-tuning. This approach keeps the base parameters unchanged while adapting to tasks through these compact layers. This method greatly improves parameter sharing, as only minimal parameters are added for each task [153].

The primary advantage of adapters lies in their ability to drastically reduce computational resources and storage requirements. Unlike traditional full-model fine-tuning, adapters only require updating a small portion of the model, thereby reducing the demand for computational power. Additionally, the limited number of parameters introduced by adapters helps reduce storage needs, making them especially suitable for environments where a model must be deployed across multiple tasks. For example, Hu *et al.* [48] introduced an LLM-adapter framework that integrates various adapters into an LLM, allowing the model to handle different tasks by fine-tuning task-specific adapters. This enabled a 7 billion (B) parameter LLM to achieve performance comparable to a 175B parameter LLM on specific tasks, demonstrating the computational efficiency of adapters. Furthermore, Rohanian *et al.* [49] demonstrated the application of adapters in a frozen BERT model to extract implicit information from clinical texts. This approach outperformed the fully fine-tuned

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BioBERT model in low-resource settings by updating only a minimal number of parameters, highlighting the efficiency of adapters in processing medical texts.

Overall, adapters offer an efficient and flexible fine-tuning strategy, particularly well-suited for scenarios where model parameters need to be shared across multiple medical tasks. By reducing both computational and storage overhead, adapters enable the efficient application of medical LLMs even in resource-constrained environments.

c) Prompt Tuning: Prompt tuning [50] is an important method of PEFT. Unlike traditional fine-tuning methods, the core idea of prompt tuning is to treat prompts as learnable parameters. Through a specialized training process, the model can generate task-specific outputs based on these learned prompts. This method updates only a small number of parameters related to the prompts, avoiding large-scale adjustments to the entire model's weights and significantly reducing computational and storage requirements. Prompt tuning includes various implementations:

i) Soft Prompts: Soft prompts [50] introduce a set of learnable embedded prompt tokens at the input, updating only these token parameters while keeping the rest of the model unchanged. For instance, Peng *et al.* [108] demonstrated that soft prompts achieve significant success in clinical concept extraction and relation extraction tasks, particularly suitable for scenarios with limited data resources or requiring rapid adaptation to new tasks.

ii) Prefix-Tuning: Prefix-tuning [51] is a PEFT technique that introduces trainable prefix vectors into each Transformer layer to adapt pre-trained model weights for specific tasks. The primary objective of Prefix-tuning is to enhance coherence and consistency in long-text generation tasks by embedding these prefix vectors directly into the attention mechanisms of every Transformer layer, offering a more comprehensive adaptation compared to methods like soft prompts, which are added only at the input layer.

Mathematically, for each Transformer layer l, Prefix-tuning introduces low-rank prefix key and value matrices $P_k^l \in \mathbb{R}^{p\times d_k}$ and $P_v^l \in \mathbb{R}^{p\times d_v}$, where p is the number of prefix tokens, and d_k and d_v are the dimensions of the key and value vectors, respectively. These prefix matrices are concatenated with the original key and value matrices $K_l \in \mathbb{R}^{n\times d_k}$ and $V_l \in \mathbb{R}^{n\times d_v}$, resulting in augmented matrices K_l' and V_l':

$$K_l' = [P_k^l; K_l] \tag{17}$$

$$V_l' = [P_v^l; V_l] \tag{18}$$

Where $[P;Q]$ denotes the concatenation of matrices P and Q along the sequence length dimension, effectively expanding the context available to the attention mechanism. This allows the model to incorporate additional task-specific information without altering the original pre-trained weights. During fine-tuning, only the prefix matrices P_k^l and P_v^l are updated, which significantly reduces the number of trainable parameters and computational overhead compared to traditional fine-tuning methods.

The optimization objective for Prefix-tuning is defined as:

$$\min_{P_k^l, P_v^l} \mathcal{L}\left(\text{Model}\left(X, \{P_k^l, P_v^l\}_{l=1}^L\right)\right) \tag{19}$$

where $\mathcal{L}$ is the loss function, X is the input data, and $\text{Model}(X, \{P_k^l, P_v^l\}_{l=1}^L)$ represents the model's output when incorporating the prefix vectors across all L Transformer layers. This approach leverages the low-rank property of P_k^l and P_v^l to efficiently capture task-specific feature variations, maintaining the integrity of the original model while enabling significant adaptability to specialized tasks.

For example, Yuan *et al.* [154] demonstrated that Prefix-tuning yielded excellent results in biomedical entity linking tasks, highlighting the method's effectiveness in enhancing the model's ability to understand and process complex medical concepts.

iii) P-tuning: P-tuning [52] extends the application of prompts by allowing learnable embeddings to be inserted at multiple positions within the input sequence, providing flexible adjustments to the model's responses for specific tasks. This approach transforms the fine-tuning process into a problem of learning optimal prompt embeddings at designated positions, thereby enhancing the model's adaptability with minimal parameter updates.

Mathematically, a set of learnable prompt embeddings $\{P^{(1)}, P^{(2)}, \ldots, P^{(k)}\}$ can be inserted at multiple positions around the input x and the label y, resulting in an augmented input X':

$$X' = \left[P_{0:i},\ x,\ P_{i+1:j},\ y,\ P_{j+1:k}\right] \tag{20}$$

In this equation, X' represents the augmented input sequence, x is the original input sequence, y is the label, $P_{0:i}$ denotes a sequence of prompt embeddings inserted before x, $P_{i+1:j}$ denotes a sequence of prompt embeddings inserted between x and y, and $P_{j+1:k}$ denotes a sequence of prompt embeddings inserted after y. By inserting multiple prompt embeddings P at various positions, P-tuning captures different aspects of the task, such as providing initial context, guiding the model's understanding between input and label, and reinforcing desired outputs.

The optimization objective for P-tuning is defined as:

$$\min_{\{P^{(j)}\}_{j=1}^k} \mathcal{L}\left(\text{Model}\left(X', \{P^{(j)}\}_{j=1}^k\right)\right) \tag{21}$$

where $\mathcal{L}$ is the loss function, and $\text{Model}(X', \{P^{(j)}\}_{j=1}^k)$ represents the model's output when processing the augmented input X' with the set of prompt embeddings $\{P^{(j)}\}_{j=1}^k$. This formulation ensures that the prompt embeddings $\{P^{(j)}\}$ are learned in a manner that minimizes the loss function while effectively guiding the model's behavior for specific medical tasks. By updating only the prompt embeddings and keeping the original model weights fixed, P-tuning significantly reduces computational and storage requirements compared to traditional fine-tuning methods.

Liu *et al.* [53] proposed an improved version of P-tuning, optimizing the learning process and parameter configuration to achieve superior performance in complex tasks such as medical text parsing. Furthermore, Liu *et al.* [54] applied the improved P-tuning to fine-tune ChatGLM-6B, achieving sig-

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nificant results in Chinese patent medicine (CPM) applications, demonstrating its potential in assisting diagnosis and treatment.

d) Instruction Tuning: Instruction tuning [55] is a PEFT technique that involves refining a pre-trained model's parameters to better align with specific task instructions. Mathematically, the instruction I is concatenated with the input X as follows:

$$X' = [I; X] \tag{22}$$

where X' is the augmented input embedding sequence with the instruction I prepended to the original input X. This technique is instrumental in scenarios requiring complex logical reasoning or multi-step tasks, significantly enhancing a model's accuracy in specified functions [55].

Song *et al.* [56] underscore the correlation between model performance and both the scale of instruction data and the size of the model, especially for tasks demanding extensive medical knowledge. Conversely, for areas like ethics where domain specificity is less critical, alternative tuning methods may be more effective.

The real-world efficacy of instruction tuning has been demonstrated in various medical applications. Yuan *et al.* [57] applied this method to the LLaMA2 13B model to develop HEAL, a model specifically for generating medical documentation, showing significant gains in quality and relevance of the output. Similarly, Wu *et al.* [58] used instruction tuning to enhance PMC-LLaMA, a model tailored for medical QA, which excelled in clinical reasoning tasks.

The optimization objective for instruction tuning is defined as:

$$\min_{I} \mathcal{L}(\text{Model}(X', I)) \tag{23}$$

where $\mathcal{L}$ is the loss function, and $\text{Model}(X', I)$ represents the model's output when processing the augmented input X' with the instruction I. This formulation ensures that the instruction embedding I is learned in a manner that minimizes the loss function while effectively guiding the model's behavior for specific medical tasks. By updating only the instruction embeddings and keeping the original model weights fixed, instruction tuning significantly reduces computational and storage requirements compared to traditional fine-tuning methods.

Liu *et al.* [53] proposed an improved version of instruction tuning, optimizing the learning process and parameter configuration to achieve superior performance in complex tasks such as medical text parsing. Furthermore, Liu *et al.* [54] applied the improved instruction tuning to fine-tune ChatGLM-6B, achieving significant results in Chinese patent medicine (CPM) applications, demonstrating its potential in assisting diagnosis and treatment. These applications illustrate instruction tuning's potential in adapting LLMs for specialized tasks, making it a critical component in the evolution of medical LLMs for diverse and complex medical scenarios.

2) External Knowledge Integration: During the optimization of LLMs, fine-tuning is a common and effective method that adjusts the internal parameters of the model to improve its performance in specific tasks. However, relying solely on the pre-trained knowledge within the model may still be insufficient for complex medical tasks, particularly when it involves the latest medical knowledge. Therefore, external knowledge integration has emerged as a key strategy to enhance the model's performance and applicability by incorporating external information sources or knowledge bases. The core idea is to fill the gaps in the model's knowledge in specific domains or tasks, allowing it to make more accurate and reliable judgments when dealing with tasks that require deep domain expertise. This approach not only supplements the model's understanding in the medical field but also helps it dynamically access real-time information, improving its ability to handle rapidly changing medical environments [155].

a) Retrieval-Augmented Generation (RAG): RAG [59] is a technique that enhances language generation models by combining parametric pre-trained models with non-parametric external knowledge sources, enabling more accurate and reliable language outputs. Unlike traditional LLMs, which primarily rely on knowledge stored within the model parameters, RAG integrates external knowledge bases to provide richer and more informed outputs.

The core of RAG lies in the integration of a pre-trained parametric model (such as LLMs) with a non-parametric memory (i.e., an external knowledge base). This non-parametric memory is accessed through a pre-trained retrieval model, such as dense passage retriever (DPR) [60], which uses a document encoder $d(z)$ and a query encoder $q(x)$ to generate dense representations for documents and queries. The inner product of these representations reflects the similarity between the document and the query:

$$p_n(z|x) \propto \exp\left(d(z)^T q(x)\right) \tag{24}$$

where $d(z)$ is the dense representation generated by the document encoder, and $q(x)$ is the query representation generated by the query encoder. These representations are typically based on the BERT model:

$$d(z) = \text{BERT}_d(z) \tag{25}$$

$$q(x) = \text{BERT}_q(x) \tag{26}$$

In practice, whenever the LLM needs to answer a question, it uses DPR to retrieve relevant content from the external knowledge base. The original input x, along with the retrieved content z, is fed into the LLM, which then leverages its powerful generative capabilities to produce precise and informative answers. This method not only improves the model's performance in specialized domains such as medicine but also significantly enhances the reliability and accuracy of its responses. For instance, Shi *et al.* [61] used RAG to extract medical facts from an external knowledge base and incorporated them into the LLM's query prompts, significantly improving the accuracy of medical QA. Ge *et al.* [62] developed LiVersa, an LLM specifically tailored for liver diseases, which integrated guidelines from the American Association for the Study of Liver Diseases (AASLD) as its RAG knowledge source, greatly enhancing the accuracy of liver-related QA and demonstrating the powerful potential of RAG to assist

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TABLE III
COMPARISON OF EXTERNAL KNOWLEDGE INTEGRATION METHODS: ADVANTAGES AND DISADVANTAGES

Method	Advantages	Disadvantages	References
RAG	1) Reduces hallucination and enhances information accuracy 2) Provides high real-time responsiveness, quickly retrieving, and generating responses 3) Offers high task adaptability, suitable for a wide range of tasks 4) Moderate explainability through provision of retrieval sources as references	1) Depends on the quality of retrieved content, may retrieve irrelevant information 2) Moderate computational resource demand 3) Requires complex implementation	[59]–[62]
KG	1) Offers strong logical reasoning capabilities 2) Ensures high knowledge accuracy based on precise structured data 3) Provides high explainability with a clear, transparent reasoning process	1) Involves high construction and maintenance costs, updates may not be timely 2) Demands high computational resources 3) Best suited for tasks that require deep reasoning, hence moderate task adaptability 4) The upper limit of real-time response speed is constrained by the query speed of the KG	[63], [64]
Tool learning	1) Extends model functionality, capable of executing complex tasks 2) High explainability with clear operational steps and results 3) Highly adaptable to diverse and complex tasks	1) Performance depends on the availability and stability of external tools 2) Moderate implementation complexity due to the need for effective integration and tool invocation 3) Real-time responsiveness is moderate, dependent on tool response speed	[65]–[68]

LLMs in generating accurate medical answers.

b) Knowledge Graph (KG): KG organizes complex data into structured knowledge networks, effectively capturing relationships among entities across various domains. In medicine, KGs structure vast data on diseases, symptoms, drugs, and treatments, enriching medical tasks with deep, structured knowledge [63]. The construction of a KG involves extracting key entities and relationships from medical texts, EHRs, and databases. It includes aligning disparate entity references into a unified representation and defining clear relationships such as "drug-treats-disease" or "symptom-indicates-disease."

Mathematically, a knowledge graph KG can be represented as a set of triples:

$$KG = \{(h, r, t) \mid h, t \in \mathcal{E}, r \in \mathcal{R}\} \tag{27}$$

where h and t are entities from the entity set $\mathcal{E}$, and r is a relation from the relation set $\mathcal{R}$. Each triple (h, r, t) represents a directed relationship from the head entity h to the tail entity t via the relation r.

Once a medical KG is constructed, the structured information from the graph can be used as additional input for training LLMs. For example, Remy *et al.* [64] developed BioLORD-2023, an LLM trained using medical KG, which demonstrated excellent performance in clinical semantic text parsing and biomedical concept representation tasks. This result highlights the significant benefits of utilizing KG data for model training.

c) Tool Learning: Tool learning is an advanced approach that enhances the functionality of language models by integrating and invoking external tools or systems. Unlike traditional methods that solely rely on built-in knowledge bases, tool learning enables models to dynamically utilize external computational resources, data systems, or specific functional modules to handle more complex tasks [65]. This is particularly important in the medical field, where information processing often requires specialized data analysis, tool manipulation, and real-time decision support [66].

The core of tool learning lies in enabling the model to automatically invoke preset external tools or APIs when performing specific tasks. For example, EHR systems and drug interaction databases are common tools. Srivastava *et al.* [67] developed an efficient data extraction tool capable of extracting data from EHRs and anonymizing it, allowing LLMs to convert structured EHR data into more easily understandable textual forms, thus enhancing their QA ability with EHR data. Bisercic *et al.* [68] introduced TEMED-LLM, a model that combines machine learning methods like decision trees and logistic regression to effectively extract structured tabular data from textual medical reports, significantly improving the accuracy of medical diagnoses.

In summary, each external knowledge integration method has its strengths and limitations in medical applications, as shown in Table III. The choice of method should depend on the specific needs and application scenarios.

However, in complex medical environments, relying on a single method might not suffice. Thus, integrating multiple methods can prove more effective. For instance, Zhu *et al.* [69] introduced the REALM framework, which synergistically combines these technologies to enhance clinical decision-making. REALM uses RAG to retrieve pertinent facts and relationships from a medical KG, aiding the LLM in generating more accurate responses by embedding this contextual knowledge into its reasoning process. Additionally, it integrates real-time data from EHRs, enriching the response relevance and timeliness. Such a comprehensive approach not only boosts the model's performance in clinical predictions but also tailors the responses to individual patient contexts, showcasing the substantial benefits of merging RAG, KG, and tool learning to elevate LLM applications in real-world medical settings.

C. How to Use Medical LLMs

In the process of using medical LLM, both prompt engineering and the mode of interaction with the model are crucial factors. The way efficient prompts are constructed to generate reasonable answers, as well as how the model is interacted with, are key determinants of the quality of the final

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result.

A prompt is an instruction given to the LLM, and its composition is critical in determining whether the model can generate high-quality results. This includes aspects such as the syntax and semantics of the prompt; factors like the order of the content in the prompt [126], and the wording of the prompt [156] can significantly influence the model's output. Prompt engineering refers to designing and optimizing input prompts to guide LLMs in generating outputs that meet specific requirements. The core of prompt engineering is constructing effective prompt sentences that enable the model to fully utilize its pre-trained knowledge to generate high-quality results. This prompt-based learning approach allows the model to perform various tasks without relying on large amounts of domain-specific data, especially in the medical field, where prompt engineering can quickly adapt to different tasks through few-shot or zero-shot learning. Transfer learning, an important application of prompt engineering, refers to the ability to apply knowledge learned from one task to another. This capability can effectively reduce redundant training and save computational resources [123]. Compared to traditional NLP models, LLMs have significantly enhanced transfer learning capabilities, making them more suitable for extensive applications in the medical field [129].

1) Hard Prompts: Hard prompts are a specific prompt technique applied to LLMs, usually consisting of direct and explicit task descriptions or examples, often requiring precise language expressions and deep domain knowledge. These prompts are typically carefully crafted by researchers to ensure the model can correctly understand and perform the specified task. For example, in a medical task, a hard prompt might be: "List the treatments related to diabetes." Such direct prompts help the model focus on specific information retrieval tasks, thereby improving the relevance and accuracy of the output. Peng *et al.* [107] utilized hard prompts to assist LLMs in extracting clinical concepts and relationships. By designing specific inquiry-based prompts, such as "Identify all mentioned drugs in the text and their effects," this method not only improved LLMs' performance in specific medical text processing tasks but also enhanced the model's ability to understand the relationships between complex medical concepts. To improve the efficiency and quality of hard prompt creation, Jiang *et al.* [70] proposed an automatic generation method for hard prompts. This method uses existing data and task descriptions to automatically construct prompts, reducing the need for manual prompt crafting while maintaining the professionalism and precision of the prompts.

2) Chain of Thought (CoT): Wei *et al.* [71] introduced the concept of CoT, referring to a process where the model produces a final output through a series of intermediate natural language reasoning steps. This method encourages the model to first engage in a sequence of internal dialogues or logical reasoning, similar to the thought process of humans solving problems. When faced with a complex problem, CoT guides the model to clarify each aspect of the problem and gradually build a reasoning path to the answer. Kojima *et al.* [72] proposed Zero-shot CoT, which employs a two-stage prompting method where the LLMs first generate preliminary reasoning through simple prompts and then generate the final answer by combining the original input with this reasoning, significantly improving reasoning performance in zero-shot scenarios. Wu *et al.* [73] evaluated the effectiveness of the CoT strategy in detecting and correcting medical errors using GPT-4, finding that this method significantly improved model accuracy. Kwon *et al.* [74] introduced a clinical CoT method requiring the LLM to provide detailed reasoning paths and explanations during disease diagnosis, which significantly enhanced the accuracy of LLMs in clinical reasoning tasks. These studies collectively demonstrate the potential of the CoT method to enhance the capabilities of LLMs, particularly in complex clinical tasks that demand high accuracy and detailed explanations.

3) Self-Consistency: Wang *et al.* [75] proposed the self-consistency method, which is based on the intuition that correct answers often exhibit higher consistency among multiple reasoning paths. This method guides LLMs to generate diverse candidate reasoning paths by providing CoT examples. The model then marginalizes these paths and aggregates the answers, ultimately selecting the most consistent answer. This unsupervised and task-agnostic method has yielded excellent results across different applications, especially in medical QA tasks. The method helps the model synthesize more accurate and consistent medical diagnoses or treatment suggestions without explicit supervision. Performance improvement is proportional to model size, and increasing the number of sampled reasoning paths can further enhance the model's ability to handle complex medical cases. Additionally, Gundabathula *et al.* [76] utilized the self-consistency method to improve LLM performance in medical error detection and correction tasks, increasing answer accuracy. Lucas *et al.* [77] proposed the ensemble reasoning method based on self-consistency, which reduces inconsistencies in medical answers and has proven effective in improving model performance in medical QA tasks.

4) Few-shot Prompts: Chen *et al.* [157] proposed the Codex few-shot prompts strategy, and Liévin *et al.* [158] applied it to medical QA tasks. This strategy involves constructing the entire context at once to derive the final answer, including a series of question-answer examples, with explanations for each answer to rule out incorrect options. However, this approach sometimes conflicts with actual clinical diagnostic processes. To better align with clinical practice, Nachane *et al.* [78] proposed the MedCodex few-shot prompts strategy tailored to the medical context. This strategy utilizes intermittent reasoning structures, answering clinical questions by building incremental contexts that simulate real diagnostic processes. The approach leverages the patient's contextual information and the model's internal knowledge for clinical reasoning, better aligning with physicians' diagnostic logic. They also found that for open-ended questions, using the MedCodex few-shot prompts strategy to first generate differential diagnoses, followed by using Codex few-shot prompts to select the final answer, achieves better results than either strategy alone, with the combination's effectiveness improving as the model size increases.

These prompt strategies can also be combined. For example,

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TABLE IV
OVERVIEW OF MEDICAL LLM EVALUATION BENCHMARKS

Name	Type	Features	Evaluation Focus	References
USMLE	Comprehensive exam	1) Highly comprehensive medical exam 2) Assesses clinical practice skills, reasoning, and communication abilities	1) Understanding and application of medical knowledge 2) Clinical decision support 3) Case analysis	[79]
MedQA	Multiple choice questions	1) Extracted from USMLE 2) Focuses on medical reasoning and decision-making	1) Diagnostic reasoning 2) Solution derivation	[80]
PubMedQA	Biomedical QA	1) Based on PubMed article snippets 2) Tests reasoning and text comprehension abilities	1) Academic literature QA 2) Information extraction	[81]
MIMIC-III	Electronic health records	1) Contains extensive ICU patient data 2) Used to assess clinical decision support and medical record summarization	1) Clinical decision support 2) Medical record summarization	[82]
MMLU-MED	Multi-task language understanding	1) Focuses on understanding and reasoning with medical knowledge 2) Sourced from medical exams	1) Medical knowledge comprehension 2) Reasoning test	[83]
MedMCQA	Multiple choice QA	1) Evaluates complex medical questions across a broad range of fields 2) Tests the ability to integrate and apply diverse medical knowledge	1) Complex medical QA 2) Diagnostic reasoning	[84]
MedAlign	EHR processing	1) Includes clinician-generated questions and instructions 2) Evaluates LLMs' performance in EHR processing	1) Clinical instruction handling 2) Longitudinal EHR analysis	[85]
MedNLI	Natural language inference	1) Evaluates inference abilities in medical text relations 2) Uses clinical notes and patient data	1) Logical reasoning in medical notes 2) Disease progression inference	[86]
MultiMedQA	Comprehensive QA	1) Combines six medical QA datasets and HealthSearchQA 2) Covers professional, research, and consumer queries	1) Truthfulness, consistency, understanding 2) Reasoning, potential harm and bias 3) Completeness, and usefulness of answers	[87]
CBLUE	Multi-task language understanding	1) A Chinese biomedical language understanding benchmark 2) Includes various tasks and real-world distribution	1) Text classification 2) Named entity recognition 3) Relation extraction	[88]

Maharjan *et al.* [159] explored the effects of integrating hard prompts, CoT, self-consistency, and few-shot prompts strategies in OpenMedLM. Without specific domain fine-tuning, the model achieved 72.6% and 81.7% accuracy on the MedQA and MMLU-MED datasets, respectively. This effectively demonstrates that combining multiple prompt strategies can significantly enhance performance in handling medical tasks.

Moreover, the way medical LLMs are used is equally important. Through remote API calls, external applications can directly interact with models deployed on the cloud or other servers, enabling functionalities like medical QA and diagnostic support. This is especially useful in scenarios requiring multi-system integration and real-time responses. On the other hand, local deployment of the model provides greater data security and customization capabilities, making it suitable for healthcare institutions with high privacy and customization needs. However, remote API calls may carry risks of privacy leakage, and local deployment can involve higher maintenance costs and complexity. Therefore, the choice of interaction method should be based on specific needs and resource availability.

IV. BENCHMARK FOR MEDICAL LLMS

Evaluation benchmarks are essential in measuring the performance of medical LLMs, particularly for tasks like clinical support and medical QA, where precise knowledge and complex reasoning are required [77]. These benchmarks not only facilitate objective assessments and comparisons across models but also help identify models' limitations in handling rare diseases and complex cases in diverse contexts [160]. By continually refining these benchmarks, researchers and clinicians can enhance the clinical utility and reliability of medical LLMs, ensuring they meet the evolving demands of patient care and medical research.

A. Overview of Medical LLMs Evaluation Benchmarks

Below are some common benchmarks for evaluating medical LLMs, summarized in Table IV:

1) USMLE: USMLE [79] is a series of highly comprehensive exams designed to assess the ability of medical practitioners in real-world practice. It evaluates not only medical knowledge but also clinical practice skills, clinical reasoning, and interpersonal skills in interacting with patients and medical teams. As a widely recognized benchmark for evaluating physicians, USMLE serves as a crucial indicator of a doctor's readiness for clinical environments. By having medical LLMs complete these exam questions, their ability to understand and apply medical knowledge, make clinical decisions, analyze complex cases, and generate reasonable diagnostic suggestions can be assessed. Additionally, USMLE includes situational questions that evaluate communication skills and response strategies in various clinical contexts, offering a unique perspective on the language generation capabilities of medical LLMs and their performance in simulated patient interactions.

2) MedQA: MedQA [80] is a multiple-choice question dataset extracted from the USMLE, focusing on medical reasoning tasks. The dataset requires models not only to recall and understand complex medical knowledge but also to derive the correct diagnosis or solution from the given question, making it a crucial benchmark for evaluating models' medi-

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cal reasoning and decision-making capabilities.

3) PubMedQA: PubMedQA [81] focuses on biomedical QA based on PubMed article snippets, requiring models to answer research-related questions. Each instance includes a question, background information, a detailed answer, and a brief yes/no/maybe answer. This benchmark tests models' abilities to process academic literature, extract key information, and generate precise answers, serving as a tool to evaluate the comprehension and reasoning skills of medical models in biomedical research contexts.

4) MIMIC-III: MIMIC-III [82] is a large, single-center intensive care unit (ICU) medical information database, containing data from patients treated in a large tertiary care hospital's ICU. The database includes vital signs, medication usage, clinician observations and notes, procedure codes, diagnosis codes, hospitalization duration, and survival data, and it is widely used to evaluate models in tasks such as clinical decision support and medical record summarization. The diverse data in MIMIC-III provides a comprehensive platform for assessing models' performance in real-world medical environments.

5) MMLU-MED: MMLU-MED [83] is a large-scale multitask language understanding benchmark aimed at evaluating models' performance in medical knowledge comprehension and reasoning. The questions typically come from medical exams or certification tests, designed to test the model's accuracy and applicability in handling specialized medical content. This benchmark provides a comprehensive framework to test models across various medical subfields.

6) MedMCQA: MedMCQA [84] is a dataset focusing on multiple-choice medical QA tasks, designed to evaluate models' performance in answering complex medical questions. This dataset covers a wide range of medical topics and serves as an effective tool for assessing models' ability to handle complex questions in real medical scenarios.

7) MedAlign: MedAlign [85] benchmark dataset is designed for evaluating LLMs in processing EHR data, featuring 983 questions and instructions from 15 clinicians. It is the first dataset to evaluate the performance of LLMs in processing clinician-generated instructions and handling comprehensive, longitudinal EHR data. Additionally, the dataset includes clinician evaluations of the LLM output, making it a key tool for testing models' ability to handle complex tasks in real clinical settings.

8) MedNLI: MedNLI [86] is a medical natural language inference dataset used to evaluate models' abilities to infer relationships within medical texts. It is primarily used to test models' performance in understanding and inferring logical relations within medical texts, such as inferring disease progression when reading clinical notes or reports.

9) MultiMedQA: MultiMedQA [87] is a comprehensive benchmark combining six existing medical QA datasets, covering professional medical, research, and consumer queries, along with a new dataset, HealthSearchQA, focused on online search medical questions. This benchmark is used to evaluate LLMs' ability to answer medical questions in exams, medical research, and consumer queries. It also introduces a human evaluation framework to assess LLMs' performance across dimensions such as truthfulness, understanding, reasoning, potential harm and bias, and the completeness and usefulness of answers, providing a holistic evaluation of LLMs' ability to apply medical knowledge in answering questions.

10) CBLUE: CBLUE [88] is a Chinese biomedical language understanding benchmark, containing multiple tasks such as text classification, named entity recognition, and relation extraction. The dataset is anonymized to protect privacy and features diversified task scenarios, following real-world distributions, thus offering a strong generalization ability. CBLUE provides a crucial tool for evaluating LLMs in the Chinese medical environment.

B. Evaluation Metrics for Medical LLMs

When evaluating medical LLMs, selecting appropriate evaluation metrics is critical. These metrics not only help measure the model's performance in specific tasks but also reveal its strengths and weaknesses in practical applications. Below are some commonly used evaluation metrics and their application scenarios in medical LLMs.

1) Accuracy: Accuracy refers to the proportion of correctly predicted samples out of all samples. The formula is as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{28}$$

where TP represents true positives, TN represents true negatives, FP represents false positives, and FN represents false negatives.

This metric is particularly valuable in medical QA and clinical information extraction, assessing the model's ability to correctly classify or respond to queries [44]. However, accuracy can be misleading in imbalanced datasets. Thus, it's often complemented by precision, recall, or F1-score to provide a balanced view of the model's effectiveness across diverse medical scenarios.

2) Precision: Precision refers to the proportion of true positive samples among the samples predicted as positive. The formula is as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{29}$$

where TP represents true positives and FP represents false positives.

Precision is vital in medical diagnostics to minimize false positives. High precision ensures that treatments and alerts are accurately targeted, crucial for conditions requiring precise identification [88]. However, overly focusing on precision might reduce the number of positive predictions, potentially overlooking true cases. Thus, it's crucial to balance precision with recall to ensure accurate and comprehensive evaluations in medical diagnostics.

3) Recall: Recall refers to the proportion of actual positive samples that are correctly predicted as positive by the model. The formula is as follows:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{30}$$

where TP represents true positives and FN represents false

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negatives.

Recall is crucial for medical screening tasks like cancer detection, where it ensures the maximum identification of potential cases, vital for early treatment and improved outcomes [85]. However, this approach can lead to overdiagnosis, prompting unnecessary medical procedures that increase healthcare costs and patient distress. Thus, while prioritizing recall, it's essential to manage false positives by balancing it with precision and specificity to ensure accurate and efficient patient care.

4) F1-Score: The F1-Score is the harmonic mean of precision and recall, balancing the two metrics. It is calculated as follows:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{31}$$

The F1-Score is a crucial metric for balancing precision and recall in medical tasks where incorrect predictions carry significant consequences. It ensures that critical conditions are captured with minimal false positives, essential for tasks like automated disease coding from EHRs [88]. However, in scenarios with extreme class imbalance, such as rare diseases, the standard F1-Score might not fully capture the model's performance, suggesting the need for a weighted F1-Score to provide a more accurate reflection across diverse conditions.

5) Domain-Specific Metrics: In addition to general classification metrics, evaluating medical LLMs involves domain-specific metrics:

a) Clinical Relevance: This metric evaluates the clinical relevance and usefulness of the content generated by the model. For example, in datasets like MIMIC-III, this metric assesses the clinical applicability of the model's output based on the evaluation by clinicians. It helps ensure that the suggestions and information provided by medical LLMs are actionable and valuable in real-world medical settings [89].

b) Reasoning Ability: This metric assesses the model's ability to perform medical logic reasoning and decision-making, including understanding and inferring logical relationships in medical text. In datasets like MedNLI, this metric measures how the model handles complex medical concepts and logical relations, evaluating its performance in advanced medical reasoning and clinical decision support [90].

C. Comparative Analysis of Existing Models

In the previous sections, we have provided a systematic overview of the lifecycle of medical LLMs and their evaluation benchmarks. To further deepen our understanding, Section IV-C presents a comparative analysis of several existing medical LLMs and general LLMs, comparing their implementations and performances. We select four widely-used medical benchmarks —MedQA, PubMedQA, MedMCQA, and MMLU-MED—to evaluate multiple models that adopt various data and fine-tuning strategies. Through these comparative case studies, we aim to identify key insights that may guide the optimization of medical LLMs and inform future research directions.

1) Model Summaries and Methods: Before presenting the benchmark results, we first provide a concise overview of representative medical LLMs, focusing on their training data and methodological choices. This background information serves as a foundation for the subsequent performance comparison.

a) Hippo-LLaMA2-7B [91]: In the get stage, it uses approximately 298 million (M) tokens of medical corpora (medical guidelines, PMC-patients, PubMedQA-contexts) for domain pre-training. In the refine stage, it applies large-scale instruction tuning with general instructions data (400 thousand samples) and evaluation instructions data, and then employs reinforcement learning for human preference alignment.

b) MEDITRON-7B/MEDITRON-70B [92]: In the get stage, it leverages 48.1B tokens of diverse medical data (clinical guidelines, paper abstracts, medical papers, replay dataset) for domain pre-training. In the refine stage, it conducts instruction tuning on multiple benchmark datasets. Additionally, MEDITRON-70B utilizes CoT and self-consistency strategies in the use stage to enhance inference quality.

c) BioMedLM-2.7B [93]: In the get stage, it employs 34.6B tokens of PubMed-related text (abstracts and full articles) for domain pre-training. In the refine stage, it designs instruction data based on downstream medical QA tasks and performs instruction tuning.

d) PMC-LLaMA-7B /PMC-LLaMA-13B [58]: In the get stage, it relies on 4.8 million medical papers and 30 000 medical textbooks (no token count specified) for domain pre-training. In the refine stage, it utilizes MedC-I (202M tokens) as instruction data for fine-tuning.

e) HuatuoGPT-II-13B [94]: It integrates the get and refine stages into a single process by converting about 5253 thousand (k) medical documents into instruction data, synthesizing 142k additional instruction samples, and completing the entire training in one unified step (no explicit token count available).

f) MMed-LLaMA 3-8B [95]: In the get stage, it pre-trains on MMedC (about 25.5B words) as a multilingual medical corpus. In the refine stage, it performs instruction tuning on medical data extracted from MMedC.

g) DRAGON-3.6B [96]: In the get stage, it fuses both general-domain (book corpus + ConceptNet KG) and biomedical-domain data (PubMed + UMLS KG) for domain pre-training. No refine or use stage enhancements are described.

h) BioMedGPT-10B [97]: Without additional pre-training, it directly enters the refine stage, applying prompt tuning on about 26B tokens of biomedical literature selected from S2ORC.

i) MING-MOE-14B [98]: Without extra pre-training, it performs LoRA-based fine-tuning in the refine stage on 300k medical samples, including 70k medical NLP task data, 130k multiple-choice questions, and 100k medical QA and dialogue data.

j) i-MedRAG-LLaMA-8B [99]: In the get stage, it adopts llama3.1-8B as the base model without further pre-training. In the refine stage, it employs iterative RAG methods during inference to dynamically retrieve medical knowledge (e.g., from textbooks and Statpearls), addressing complex medical QA tasks.

k) Apollo-2B [100]: In the get stage, it uses 2.5B tokens of

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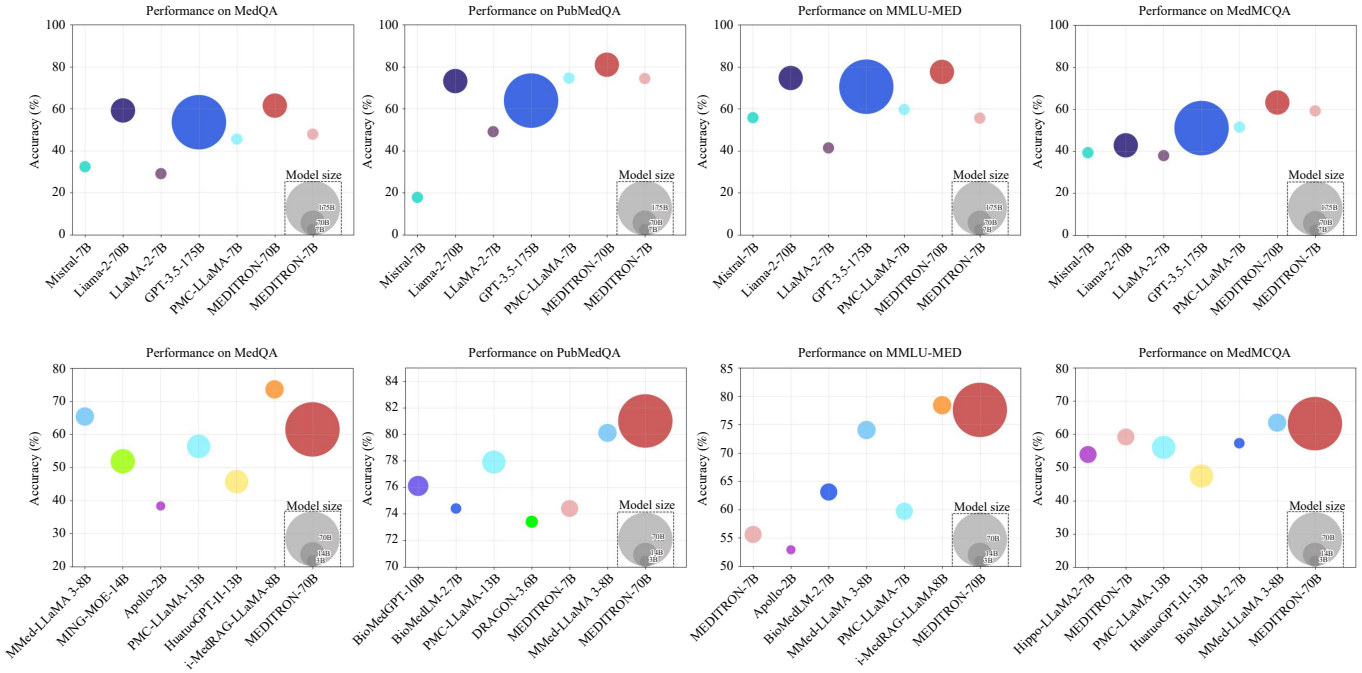


Fig. 2. Model performance comparison on different datasets. The top row presents a comparison of accuracy between general LLMs and medical LLMs across the MedQA, PubMedQA, MMLU-MED, and MedMCQA datasets. The results demonstrate that medical LLMs generally achieve higher accuracy, highlighting their advantages. The bottom row compares the accuracy of medical LLMs employing different methodologies on the same datasets, indicating that the impact of varying methods on performance can even surpass differences in model parameter sizes.

multilingual medical data from ApolloCorpora (including books, encyclopedias, papers, exams, etc.) for domain pre-training. In the refine stage, it converts the original corpus into QA pairs to achieve instruction tuning.

2) Performance on Benchmark Datasets and Insights: We evaluated the accuracy of these models on MedQA, PubMedQA, MMLU-MED, and MedMCQA as shown in Fig. 2. The top row of Fig. 2 compares the accuracy of general LLMs and medical LLMs across these datasets, demonstrating that medical LLMs generally achieve higher accuracy, highlighting their advantages. For instance, general LLMs typically underperform compared to medical LLMs, with GPT-3.5 (175B parameters) showing inferior performance on most medical datasets compared to LLaMA-2-70B. This suggests that as general capabilities increase, domain-specific abilities may diminish. Additionally, both LLaMA-2-70B and LLaMA-2-7B generally lag behind their medical counterparts of similar parameter sizes, and in some cases, smaller medical LLMs even outperform larger ones on certain datasets.

The bottom row of Fig. 2 compares the accuracy of medical LLMs employing different methodologies on the same datasets. Our experiments demonstrate that techniques such as RAG enable smaller models to outperform larger ones by effectively integrating external medical knowledge, thereby compensating for fewer parameters and substantially improving answer accuracy. For example, MEDITRON-7B significantly underperforms MEDITRON-70B, indicating that increasing model parameters often contributes to better performance. However, i-MedRAG-LLaMA-8B surpasses MEDITRON-70B, highlighting the considerable value of RAG in medical QA tasks.

Further examination reveals that BioMedLM-2.7B outperforms some 7B-parameter models in PubMedQA, MMLU-MED, and MedMCQA, indicating that high-quality and large-scale data play a crucial role in boosting model performance during both the pre-training and fine-tuning stages. Nonetheless, a single model may exhibit inconsistent performance across different benchmarks. For example, MEDITRON-7B falls far behind BioMedLM-2.7B in MMLU-MED but does not show a notable gap in PubMedQA and MedMCQA. This discrepancy reflects the limitations of current evaluation benchmarks: each dataset's distinct focus and difficulty prevent a comprehensive assessment of a model's true capabilities.

Regarding the LoRA-based MING-MOE-14B, its performance in MedQA is not substantially weaker than that of other models with comparable parameter scales. This suggests that LoRA can effectively reduce training costs in resource-constrained scenarios and mitigate the need for extensive parameter adjustments while maintaining satisfactory performance. Although most models follow a "pre-train and instruction tuning" paradigm, their performance still varies considerably. MMed-LLaMA 3-8B, for instance, consistently performs well across all four benchmarks, nearly matching MEDITRON-70B. This advantage likely stems from the richness and high quality of the MMedC dataset (25.5B tokens of multilingual medical corpora). In contrast, PMC-LLaMA-13B and HuatuoGPT-II-13B, despite leveraging large amounts of medical literature, fail to achieve expected performance due to insufficient data quality, diversity, or scale.

In summary, these comparisons suggest that beyond model size, the quality, diversity, and domain-specific relevance of both pre-training and fine-tuning data are critical to enhancing the performance of medical LLMs. Even a massive train-

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ing set (e.g., MEDITRON's 48.1B tokens from various sources) may not guarantee optimal results without linguistic diversity and high-quality data. Moreover, methods like RAG exhibit considerable potential—smaller-parameter models can still outperform larger models by retrieving external domain knowledge. Collectively, these findings imply that rather than solely increasing parameter counts, directing greater effort toward data quality control, linguistic diversity, domain adaptation, and the integration of external knowledge could be more impactful for advancing the state of medical LLMs.

D. Limitations of Existing Benchmarks

Despite the important role that existing medical LLM evaluation benchmarks, such as USMLE, MedQA, and MIMIC-III, play in assessing medical knowledge, reasoning abilities, and clinical applications, they also have significant limitations.

1) Insufficient Complexity of Scenarios: Current benchmarks, such as MedQA and MedNLI, primarily focus on assessing fundamental medical knowledge and logical consistency, but they often fall short when it comes to handling complex clinical scenarios. These benchmarks typically adopt multiple-choice or static text-based QA formats, lacking the ability to simulate dynamic clinical decision-making environments. As a result, they fail to fully capture the model's real capabilities in handling multi-step decision-making, changes in patient conditions, or multidisciplinary collaboration. Krithara *et al.* [101] point out that these benchmarks often fail to meet the actual informational needs of medical professionals, providing only basic answers with limited depth. Lyu *et al.* [102] highlight that existing datasets oversimplify clinical evidence reasoning questions, falling short of addressing the needs of real-world clinical environments.

2) Lack of Coverage of Real-World Data: Most benchmarks are based on structured data or carefully curated texts, limiting their ability to evaluate models on real-world clinical data. For instance, although the MIMIC-III dataset provides rich clinical data, it still falls short in terms of data diversity and representativeness, particularly when dealing with patient data from different healthcare systems or cultural backgrounds. Luo *et al.* [103] mention that many benchmarks lack sufficient coverage of question types involving real-world tabular data. Similarly, Nguyen *et al.* [104] argue that current datasets do not sufficiently include consumer health-related questions, which often present answers that are difficult to understand and fail to meet practical needs.

3) Challenges in Evaluating Open-Ended Questions: When evaluating open-ended questions, existing benchmarks often rely on semi-automated human-model collaboration, which increases the complexity and cost of evaluation. Common evaluation metrics such as accuracy or F1-Score are often inadequate for precisely quantifying the similarity between model-generated answers and standard answers. Wang *et al.* [105] highlight that current benchmarks oversimplify the evaluation of open-ended questions, making it difficult to comprehensively compare the performance of different models. Johnson *et al.* [82] mention that different prompts can lead to inconsistent model outputs, complicating the evaluation of open-ended questions and making it difficult to accurately assess the model's actual capabilities.

4) Lack of Consideration for Ethical and Privacy Issues: Current evaluation benchmarks often overlook the model's performance in ethical decision-making and patient privacy protection. As the use of models in the medical field grows, it becomes increasingly important to assess these aspects. For example, Mediboina *et al.* [106] suggest that large medical models need enhanced ethical safeguards when handling sensitive health information. However, existing benchmarks do not sufficiently cover the evaluation of such safeguards, limiting the comprehensiveness and practical utility of model assessments.

5) Linguistic and Cultural Diversity: Current evaluation benchmarks are predominantly focused on English-language environments, which limits their applicability in non-English medical settings. While benchmarks like CBLUE have filled some gaps for Chinese, there remains a lack of benchmarks for other languages and cultural contexts, restricting the broader global application of medical large models. Additionally, the diversity of languages and cultures across different regions requires models to possess greater adaptability and sensitivity to ensure efficient and accurate medical decision support across various cultural and linguistic backgrounds. Developing more multilingual evaluation benchmarks, such as for Arabic, Hindi, and Spanish medical QA and diagnostic tasks, is a key direction for future benchmark development, which will improve the applicability and effectiveness of models in global healthcare environments [100].

V. Medical Applications

In this section, we will explore the specific applications of medical LLMs in various domains, focusing on their role and impact in clinical practice, research activities, and medical education. Through an analysis of these application scenarios, we will gain a deeper understanding of how medical LLMs are driving the development of the healthcare industry and explore their potential for future applications in broader fields.

A. Clinical Applications

1) Clinical Information Extraction: Clinical information extraction is a critical task that involves automatically identifying and extracting medical knowledge from clinical texts. This process typically includes two main steps: concept extraction and relation extraction. Concept extraction involves identifying and extracting specific medical terms or entities from the text, such as diseases or symptoms; relation extraction refers to establishing semantic relationships between concepts, such as what symptoms are associated with a disease [161]. Dai *et al.* [162] emphasize the importance of extracting family history information from clinical notes, as this information is crucial for predicting hereditary diseases and formulating preventive health measures. Effective clinical information extraction not only enhances data usability but also significantly improves the accuracy and efficiency of clinical decision-making.

LLMs are well-suited for tasks involving the identification of clinical concepts and relationships from clinical texts. machine reading comprehension (MRC) architectures are particularly adept at these tasks [163]. Peng *et al.* [107] devel-

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oped an MRC architecture based on hard prompts to handle clinical concept and relation extraction, identifying concepts and relationships by answering questions. This method uses hard prompts as inputs, capturing task-related concepts and relationships to guide the model in completing the task correctly. These prompts rely on the prior knowledge of medical experts but avoid the computational cost of training a model, showing strong performance in clinical concept and relation extraction tasks. Peng *et al.* [108] further developed an MRC architecture based on soft prompts to replace hard prompts, using trainable soft prompts with LLMs for clinical concept and relation extraction. This approach demonstrated good transfer learning and few-shot learning capabilities, proving the feasibility of applying soft prompts to medical LLMs. Peng *et al.* [109] added soft prompts to both the input layer and all transformer layers for clinical concept and relation extraction, achieving good results across institutions and disease domains with strong transfer learning capabilities.

2) Medical Question Answering: Using medical LLMs to answer health-related questions not only provides accurate medical information and advice but also improves the efficiency of healthcare services and empowers patients to manage their health better. Lai *et al.* [110] developed the Psy-LLM framework, integrating pretrained LLMs with psychological QA and articles to enhance mental health support. They evaluated the model's content generation capabilities related to psychological counseling through intrinsic metrics and human assessments, demonstrating its potential to significantly boost healthcare efficiency.

To overcome the limitations of general LLMs in medical contexts and reduce factual errors during patient interactions, Li *et al.* [7] created ChatDoctor. This medical QA model enhances the LLaMA-7B by employing instruction tuning and integrating RAG techniques. It was fine-tuned using a dataset of 100 000 patient-doctor dialogues from an online medical consultation site, enriched with external knowledge from disease databases and Wikipedia, thereby improving its ability to understand patient needs and provide accurate advice. Addressing issues of hallucination and bias in medical QA, Wang *et al.* [111] introduced the joint medical LLM and retrieval training (JMLR) method. This approach connects LLMs with retrieval systems by using the relevance score of LLM-generated answers to RAG-retrieved documents as the retrieval system's training loss. This innovative training method allows simultaneous optimization of both the LLM and the RAG retriever, significantly enhancing performance in medical QA and reasoning tasks, reducing hallucinations, and boosting the model's interpretability and reliability.

In the era of globalization, the demand for medical services transcends linguistic and cultural barriers. As medical LLMs strive to serve diverse patient populations, addressing the multi-language and multi-cultural aspects becomes crucial. Different languages carry unique medical terminologies, idiomatic expressions, and cultural connotations related to health and illness. To popularize medical LLMs among people in more regions, Wang *et al.* [100] proposed Apollo, a medical question-answering LLM that covers six languages including French, Spanish, and Arabic. This initiative not only breaks down language barriers for medical workers who are not native English speakers but also takes a significant step towards providing more inclusive and culturally sensitive medical services. By supporting multiple languages, Apollo can better adapt to the diverse medical information-seeking behaviors and communication styles of different regions, ensuring that medical knowledge and advice can be effectively conveyed across different linguistic and cultural backgrounds.

3) Medical Document Processing: In the medical field, medical document processing utilizes medical LLMs to automate tasks and improve accuracy. These models can efficiently generate structured medical documents from complex and abbreviated data without the need for explicit annotations, greatly simplifying the medical operation process.

Waisberg *et al.* [120] demonstrated the excellent ability of GPT-4 in drafting discharge summaries for patients after cholecystectomy, optimizing the discharge process. Similarly, Clough *et al.* [113] found that ChatGPT can assist doctors in creating high-quality, structured discharge summaries, saving time and reducing the administrative burden. Cascella *et al.* [112] showcased ChatGPT's capability to generate structured medical summaries from abbreviated ICU patient data and accurately categorize the information, providing support for medical decision-making.

Privacy issues are of vital importance in medical document processing. Wiest *et al.* [164] developed an LLM-based anonymization tool, which can enable on-site de-identification of data and reduce the problem of privacy leakage in medical documents. Differential privacy is a mathematical framework that safeguards data privacy during the process of data analysis by adding calibrated noise [165]. While protecting patients' privacy, it still enables valuable statistical analysis to be carried out. Federated learning, on the other hand, is a distributed approach that allows multiple parties (such as hospitals) to collaboratively train models without sharing their raw data [166]. Song *et al.* [167] proposed a new method that transforms the task of generating differential privacy-based pseudo-samples into a discrimination task. This method combines a large language model and a discriminator for text classification in the private domain, balancing privacy and practicality in medical documents. Liu *et al.* [168] proposed a federated learning method called DP-LoRA, which protects data privacy by adding noise to weight updates using the Gaussian mechanism, promoting collaborative model training while maintaining data privacy in medical documents.

B. Research Applications

Clinical trials are crucial for assessing new medical interventions but are often hindered by high costs, lengthy processes, and low success rates [169]. One notable innovation, the TWIN-GPT model developed by Wang *et al.* [114], utilizes the extensive clinical knowledge embedded in LLMs alongside real-world data to create personalized digital twins. This model effectively simulates individual physiological conditions and treatment outcomes, even in scenarios with limited data. Experimental results show that TWIN-GPT not only enhances the accuracy of clinical trial predictions but also pro-

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duces high-fidelity data that mimics specific patient conditions, significantly improving outcomes in research environments where data are scarce. This breakthrough underscores the transformative potential of integrating LLMs in medical research, particularly in optimizing clinical trials and personalizing patient care.

Additionally, Tayebi Arasteh *et al.* [115] explored the application of ChatGPT's advanced data analysis (ADA) in automating machine learning (ML) analyses of clinical trial data. This application bridges the gap between ML specialists and clinicians, facilitating the adoption of ML in clinical research. By applying ADA to datasets from diverse clinical trials —ranging from endocrine tumors to cardiac amyloidosis —ChatGPT demonstrated its capability to autonomously develop and refine ML models. These models matched or exceeded the performance of traditional approaches, illustrating that LLMs can empower clinicians to not only design but also optimize ML models, thereby enhancing the depth and breadth of medical research.

C. Educational Applications

In medical education, the rise of digital resources has shifted the focus from rote memorization to critical thinking and innovation, increasing demands on educators. LLMs facilitate this transition by providing personalized content delivery, condensing complex medical information, and enhancing the learning experience [170], [171].

Seetharaman [116] shows how ChatGPT can offer language support to these students, providing feedback on English writing style and language usage, helping them improve their expression of medical knowledge. Birkun and Gautam [118] highlight that LLMs can support community resuscitation education and provide necessary information to non-professional rescuers in emergencies. Additionally, Ahn [119] demonstrated the potential of LLMs to deliver cardiopulmonary resuscitation (CPR) knowledge quickly and accurately to the public through ChatGPT. Hasnain [172] evaluated ChatGPT's effectiveness in disseminating information on monkeypox symptoms and prevention measures, highlighting its value as an educational resource for both the public and medical students.

Villagrán *et al.* [117] have shown that LLMs can reduce the educational burden on instructors by automating feedback and facilitating cognitive rehearsal in disciplines like physical therapy. This integration into existing educational frameworks has shown to enhance student understanding and engagement.

Waisberg *et al.* [40] examined the feasibility of using GPT-4 for educational purposes. They found that GPT-4 can provide customized learning experiences tailored to students' needs and abilities, increasing interactivity and making the learning process more engaging.

VI. Challenges and Future Prospects

A. Challenges

Despite the significant potential demonstrated by medical LLMs in clinical, research, and educational applications, they still face numerous challenges.

1) Unfathomable Datasets: "Unfathomable datasets" highlight the challenges posed by the vast and complex datasets used to train medical LLMs, where their immense scale makes thorough human verification impractical. This vastness, encompassing extensive patient records and clinical trial data, critically impacts the quality and reliability of model training [125]. The sheer volume and detail in these datasets render traditional quality assurance methods inadequate, potentially leading to data redundancy and inefficiencies in model learning [17]. Consequently, these factors heighten the risk of incorrect medical diagnoses and recommendations, posing significant challenges to the deployment and development of medical technologies.

2) Limited Context: "Limited context" in medical LLMs describes their inability to fully utilize or remember past interactions during extended dialogues, critically hindering their capability to manage complex medical scenarios requiring detailed historical data. This issue is mainly due to the fixed context window size of these models and their training data constraints, which limit their capacity to integrate lengthy medical histories or diverse information sources effectively [127]. This limitation is particularly evident in models like GPT, which struggle to synthesize extensive details from varied time points and data types, leading to potential inaccuracies in diagnostics and treatment recommendations [173]. Consequently, this can reduce the trust in LLM technology among medical professionals and patients, impacting the practical utility of LLMs in healthcare settings.

3) High Computational Demand: "High computational demand" refers to the significant resources needed for medical LLMs to process data and perform tasks effectively. This stems from the complexity of these models and the healthcare sector's need for precision and immediate processing, necessitating advanced hardware like GPUs or TPUs. Patterson *et al.* [124] highlight the environmental and financial costs, noting that training LLMs can generate carbon emissions comparable to those of a large city over a month, with expenses reaching into the billions. Such demands can limit the deployment of medical LLMs, particularly in resource-poor areas, worsening global healthcare disparities [174]. This issue challenges the sustainability and equity of medical LLMs, impacting their global accessibility and acceptability.

4)Lack of Interpretability: "Lack of interpretability" in medical LLMs refers to the difficulty in understanding how these models make decisions, a significant barrier in healthcare settings. This "black box" effect prevents healthcare professionals from fully trusting and validating the model's recommendations, potentially compromising patient safety [123]. The complexity of these systems and the absence of a unified framework for interpretability mean that even expert clinicians may struggle to understand the rationale behind specific model outputs, which complicates the adoption of LLMs in clinical practice [175]. Furthermore, the inability to elucidate decision-making processes can lead to legal and ethical issues when medical incidents occur [176]. Concerns about the models providing unverified or plagiarized content add another layer of risk, increasing the liability associated with their use

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[177].

5) Knowledge Update Lag: "Knowledge update lag" in medical LLMs highlights the difficulty of incorporating the latest medical findings into models swiftly. As medical knowledge continuously evolves, keeping LLMs current requires frequent updates, which can be expensive and technically challenging [125]. Hoelscher-Obermaier *et al.* [178] illustrate that updating models to discard obsolete or erroneous information without affecting related accurate data can introduce new errors, complicating the update process. This lag not only risks using outdated information, potentially leading to poor medical advice, but also affects the trust and reliability perceived by healthcare professionals, limiting the practical utility and advancement of medical LLMs in research and clinical practice.

6) Hallucination: "Hallucination" in medical LLMs refers to the generation of plausible but incorrect information, posing significant risks in clinical settings where accuracy is paramount. This challenge often arises from the model's training on inadequate or biased data, leading to errors when generalizing from underrepresented scenarios. For example, Cascella *et al.* [112] noted that ChatGPT could produce convincing yet inaccurate medical advice, underscoring the necessity for rigorous verification processes. Similarly, Paullada *et al.* [121] observed hallucinations in scenarios involving rare diseases, where the model extrapolated beyond its training data, leading to potentially harmful recommendations. Furthermore, Chen *et al.* [179] reported that during stress tests, LLMs often fabricated symptoms or treatment details, complicating the clinical decision-making process. These issues highlight the critical need for enhancing the reliability of LLMs in healthcare, ensuring they augment rather than compromise patient care.

7) Ethics: "Ethics" in medical LLMs is intricate and multifaceted, with implications that extend far beyond the realm of technical development. At their core, these considerations revolve around ensuring privacy, fairness, and the proper handling of sensitive patient data. However, the path to achieving these goals is fraught with challenges, as issues such as data breaches, bias, and the misuse of patient information pose significant ethical conundrums [122], [180]. Patients are increasingly aware of the sensitive nature of their medical information, which can include details about their genetic makeup, chronic conditions, and personal health histories. When patients decline to give consent, it creates a significant hurdle for model development. Moreover, any attempt to use patient data without proper consent may result in serious legal consequences [164]. The lack of diverse data due to non-consent can lead to incomplete models, unable to accurately represent the full spectrum of patient demographics, medical conditions, and treatment. Bias is another major ethical concern in medical LLMs. As Paullada *et al.* [121] have highlighted, training datasets often contain biases that can lead to discriminatory outcomes. These biases can stem from various sources, including the over-representation of certain patient groups in the data, historical biases in medical research, or inaccuracies in data collection.

8) Evaluation Challenges: "Evaluation challenges" for medical LLMs arise from the complex nature of medical data and the lack of unified standards for assessment [101], [103], [105]. This complexity is exacerbated when models must interpret diverse data types or answer open-ended questions, making traditional metrics inadequate. These evaluation difficulties can undermine the trust of healthcare professionals in the reliability of LLM-generated advice, limiting their use in practice [102], [104]. Furthermore, Lu *et al.* [126] demonstrate how variations in prompts can significantly impact response quality, underscoring the need for more robust and comprehensive benchmarks to ensure the evaluations do not overstate the models' capabilities.

B. Future Directions

As technology continues to advance, future developments in medical LLMs will increasingly focus on improving accuracy and transparency. The integration of new techniques such as reinforcement learning [181], model editing [182], and synthetic data [183] will enhance LLMs' ability to self-correct and adapt, making them more aligned with human values. Moreover, the development of multimodal LLMs capable of handling various data types will broaden their scope of application. These innovations are expected to drive the widespread adoption of medical LLMs in the global healthcare sector, significantly improving the quality and efficiency of medical services.

1) Reinforcement Learning: Reinforcement learning [181] is a commonly used method for aligning medical LLMs with human values. It is a learning technique based on interactions with an environment, where feedback is used to improve the model's performance. One of its key applications in LLMs is to help the model better align with human values and preferences when generating content, which is especially crucial in the medical field. While traditional fine-tuning methods can adapt models to specific tasks, they often overlook human ethical principles and social values. In sensitive medical applications, incorrect outputs can have serious consequences. By incorporating reinforcement learning, we can fine-tune the model's behavior to ensure that its answers are more aligned with human expectations and ethical standards [184], [185].

The core idea of reinforcement learning is to interactively adjust the model's output strategies based on human feedback, maximizing the score from human input. A commonly used approach is reinforcement learning from human feedback (RLHF) [186], which helps LLMs generate answers that align with human preferences [187]. The primary mechanism involves training a reward model based on human feedback on the LLM's responses, using it to evaluate whether the generated text meets human needs, and optimizing the LLM using policy gradient algorithms, such as proximal policy optimization (PPO) [188], to align the LLM with the reward model.

Another approach is direct preference optimization (DPO) [189], which simplifies the standard RLHF problem by addressing it with a straightforward classification loss, without the need for training and fitting a separate reward model.

Reinforcement learning holds great potential in the field of medical LLMs. Take the emerging reasoning model DeepSeek R1 [190] as an example. It employs reinforcement learning

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methods and demonstrates a powerful ability to solve complex reasoning problems. This is extremely beneficial for the reasoning of complex diseases in the medical field. It enables the model to effectively simulate doctors' diagnostic thinking and thus provide more accurate diagnoses.

Zhou *et al.* [191] proposed a method that achieves RLHF-level results by fine-tuning the 65B LLaMa model with only 1000 carefully selected prompt-response pairs, using standard supervised loss. This demonstrates that LLMs learn most knowledge during pre-training, and only need instruction fine-tuning to generate high-quality outputs that meet human needs.

These methods enable reinforcement learning to significantly improve the ethical alignment of medical LLMs when generating content, ensuring safer and more reliable outcomes in real-world applications.

2) Model Editing: Model editing [182] is a technique used to modify trained LLMs to correct errors or adjust outputs to meet specific requirements or standards. This technique allows for fine-tuning of the model's parameters without the need to retrain the entire model from scratch, thereby saving resources and enhancing efficiency [192]. Below, we introduce some common methods for model editing.

MEND [182] fine-tunes a pre-trained language model by training a secondary model known as an editor, which proposes parameter changes based on specific editing tasks. This approach enables the model to quickly adapt to new data or correct systemic errors without extensive retraining. ROME [193] enhances the concept of model editing by introducing a framework that not only applies edits but also optimizes these edits through iterative refinement. This method focuses on making more targeted and effective modifications to the model, improving the precision of the edits while ensuring that changes do not adversely affect the model's performance on the originally trained tasks. MEMIT [194] is a method that combines model editing with model interpretability techniques to gain a clearer understanding of why certain edits are effective or necessary. This method employs interpretability tools to analyze the impacts of potential edits before applying them, aiding in more accurately guiding the editing process and ensuring that changes align with expected outcomes. Additionally, Gangadhar and Stratos [195] have demonstrated that fine-tuning is also a viable method for model editing, allowing for adjustments while preserving the model's core learning.

In medical LLMs, model editing can play a crucial role. For example, when a medical LLM generates inconsistent summaries of medical records or inaccurate medical information in patient consultation responses, model editing can directly modify specific parts of the model, leading to more accurate and reliable outputs [196]. This not only enhances the model's performance in practical medical applications but also builds trust among healthcare professionals and patients in the technology [197].

Moreover, model editing helps in rapidly adapting to the latest developments in the medical field, such as new treatment methods or drug information, ensuring that model outputs reflect the most current medical knowledge and practice standards. This is extremely valuable in the dynamically changing healthcare environment, effectively supporting healthcare professionals in making better diagnostic and treatment decisions [198].

3) Synthetic Data: Synthetic data techniques [183] enhance the scale and diversity of datasets by creating new data, which is particularly crucial in situations where data is insufficient or unevenly distributed. This is essential for improving the generalization capabilities and task performance of medical LLMs.

On the one hand, synthetic data can be used to generate new data. High-quality medical data is crucial for training medical LLMs, but due to privacy concerns, high annotation costs, and limited real-world diversity [199], it is challenging to obtain large volumes of high-quality, annotated medical data. Synthetic data is a cutting-edge solution to this issue. A popular method involves using medical LLMs to create high-quality medical data, thus improving the effectiveness of model training and data diversity. For instance, Tang *et al.* [200] developed a new framework that uses ChatGPT to generate high-quality synthetic data, which was then used to train local models. This significantly improved model performance and reduced the time and effort required for data collection and annotation. Brekke *et al.* [201] proposed an iterative approach to annotate and extract family history information from clinical texts to generate synthetic data, and evaluated models trained with both synthetic and real clinical data. The results showed minimal differences, demonstrating the superior performance of synthetic data in model training.

On the other hand, synthetic data can also be used to balance class proportions within datasets, reducing cultural or racial biases. For instance, in original datasets, some cultural or racial groups may be underrepresented, leading to poorer model performance on tasks related to those groups, thus contributing to cultural bias. Moreover, these datasets may contain discriminatory features, which could cause the trained LLM to produce biased outputs or discriminatory language, leading to racial bias [202]. By generating more data related to these groups, synthetic data can balance the distribution and allow the model to learn more comprehensive and unbiased knowledge, reducing the risk of bias [114]. For example, Pezoulas *et al.* [203] proposed a synthetic data method that simulates virtual profiles for underrepresented populations, effectively addressing imbalanced data distribution issues. Additionally, synthetic data can help mitigate privacy concerns by generating de-identified synthetic data from personal sensitive information, which helps protect patient privacy in clinical texts [204]–[206]. For instance, Guillaudeux *et al.* [207] proposed a synthetic data approach that generates random "avatar data" centered on the patient, using distance-based privacy measures to preserve key clinical information while removing sensitive patient details.

Synthetic data not only expands the scale of datasets but also enhances the diversity and representativeness of the data, significantly improving the training effectiveness and model performance of medical LLMs. It is expected that, in the

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future, synthetic data will play a crucial role in the development of medical LLMs, providing broader and more balanced data support for model training, thereby driving technological advancements in the healthcare field.

4) Multimodal LLMs: Multimodal LLMs refer to LLMs capable of simultaneously processing and understanding information from multiple data sources, including text, images, and audio. By integrating data from different modalities, these models can capture and understand complex information scenarios more comprehensively, thereby improving the accuracy of processing and predictions. In the medical field, this provides significant advantages; for instance, text information provided by patients can describe their symptoms, ask questions, and summarize medical records [78]. Audio from patients can assist in diagnosing conditions such as respiratory diseases and Parkinson's disease, which are closely related to speech patterns [2]. Additionally, image data from patients can reveal pathological changes and the progression of diseases [208]. Thus, applying multimodal LLMs in healthcare is highly beneficial [209]. However, there is still substantial room for growth in this domain. Building on the work of Huang *et al.* [210] and their OpenPath dataset, future research could focus on developing more sophisticated algorithms to better fuse the pathological images and natural language descriptions. For example, exploring ways to enhance the pathology language-image pretraining (PLIP) model's ability to capture the intricate relationships between visual features in pathological images and semantic information in the text, enabling more accurate and nuanced medical assessments.

In line with the creation of the PubMedVision dataset by Chen *et al.* [94], another critical future direction is to further integrate text with other modalities, such as audio data. Given that audio can provide valuable diagnostic clues for certain diseases [2], developing multimodal LLMs that can seamlessly combine textual medical knowledge, visual information from medical images, and audio data could significantly improve diagnostic accuracy. For instance, models could be trained to analyze a patient's spoken symptoms (audio), relevant medical images, and textual records simultaneously, enabling more comprehensive and precise diagnoses.

Another crucial area for future exploration is the development of medical LLMs that can adapt to evolving medical knowledge in real-time. As new medical research emerges constantly, these models should be designed to update their knowledge bases promptly. Drawing inspiration from the work of Oh *et al.* [211] and their integration of updated clinical text information into the clinical target volume (CTV) delineation model, future multimodal LLMs could be enhanced to not only incorporate new textual medical knowledge but also integrate the latest findings from image-based and audio-based medical research. This real-time adaptability will ensure that the models can provide up-to-date and reliable medical advice.

In addition, with the increasing importance of vision-language models in healthcare, further analysis of their unique capabilities and potential applications is warranted. Vision-language models, a subset of multimodal LLMs, have the potential to revolutionize medical image interpretation. Future work could build on the achievements of HuatuoGPT Vision [94] and PLIP [210] to optimize these models to better understand complex medical images, such as X-rays, MRIs, and CT scans, and combine this understanding with textual medical knowledge. This could lead to more accurate diagnoses, better treatment planning, and ultimately, improved patient outcomes.

These examples illustrate the vast potential of multimodal LLMs in medicine. Future research can fully leverage their capabilities to enhance the quality of healthcare services, reduce medical costs, and improve patient experiences.

VII. Conclusion

LLMs have demonstrated immense potential in the medical field, offering a promising solution to address global disparities in healthcare resources. In this paper, we provide a detailed overview of the lifecycle of medical LLMs, allowing readers to gain a comprehensive understanding of the technologies involved. We also introduce commonly used benchmarks and metrics to facilitate the evaluation of medical LLM capabilities. Furthermore, we discuss the specific applications of medical LLMs in the healthcare domain and the challenges they face, helping readers grasp their practical significance. To address these challenges, we introduce several potential methods for enhancing model performance, which could significantly improve the effectiveness of medical LLMs in the future. We believe that as these technologies continue to evolve and mature, medical LLMs will play a pivotal role in advancing the future of healthcare.

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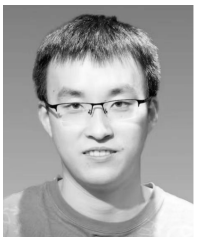


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